

Algorithms HW

1. A *circular polygon* $\mathcal{C}$ is a simply connected open set of $\mathbb{C}$ whose boundary is a union of circular arcs and is homeomorphic to a circle. Show that any Riemann map $\Delta \rightarrow \mathcal{C}$ extends to a homeomorphism $\overline{\Delta} \rightarrow \overline{\mathcal{C}}$.

■

2. Let $f : \Delta \rightarrow \Delta$ be a holomorphic function that extends continuously to $\bar{\Delta} \rightarrow \bar{\Delta}$, with $f(S^1) \subset S^1$. Show that f can be extended to a holomorphic function on $\mathbb{C}$ with finitely many points removed.

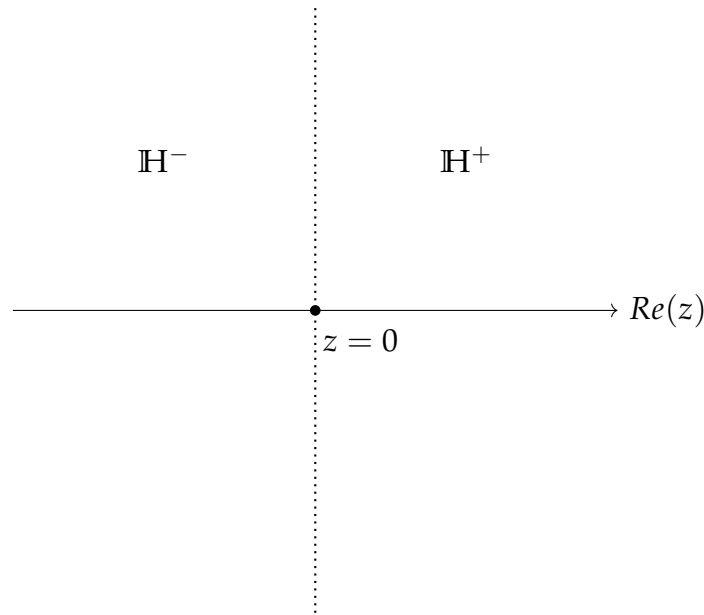
Recall that we have a biholomorphism from the $\Delta \rightarrow \mathbb{H}$ given by

$$h(z) = i \cdot \frac{1+z}{1-z}.$$

This map extends continuously to the boundaries $\bar{\Delta} \rightarrow \bar{\mathbb{H}} \cup \infty$ with $h(1) = \infty$. And so we can define a holomorphic function $g : \mathbb{H} \rightarrow \mathbb{H}$ by $g = h \circ f \circ h^{-1}$ which extends continuously to the actual boundary $\bar{\mathbb{H}} \cup \infty$ and $g(\bar{\mathbb{H}} \cup \infty) \subseteq \bar{\mathbb{H}} \cup \infty$. The preimages of 1 under f are those which get sent to infinity under g .

Suppose for a moment that we have a single point on the real line which gets sent to infinity. Say that we translate our coordinates so that this point is $z = 0$. Let us denote the “right half upper plane” by $\mathbb{H}^+ := \{z \in \mathbb{H} : \operatorname{Re}(z) > 0\}$ and similarly let $\mathbb{H}^-$ denote the “left upper half plane”.

Now, the restriction of g to $\mathbb{H}^+$ defines a holomorphic function which extends continuously to the interval $(0, \infty)$ with the image of $(0, \infty) \subseteq \mathbb{R}$. The Schwarz reflection principle then allows us to extend $g|_{\mathbb{H}^+}$ to a holomorphic function on the entire right half plane $\mathbb{R}^+ := \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$. The same argument means that we can extend $g|_{\mathbb{H}^-}$ to a holomorphic function on the entire left half plane. We now have a holomorphic function $\mathbb{C} \setminus L \rightarrow \mathbb{C}$ where L is the vertical line through $z = 0$. Let us denote this function by $\tilde{g}$.



We can extend $\tilde{g}$ further to a function $G : \mathbb{C} \setminus \{z = 0\} \rightarrow \mathbb{C}$ in the following way

$$G(z) = \begin{cases} \tilde{g}(z) & z \notin L \\ g(z) & z \in L \cap \mathbb{H} \\ \overline{g(\bar{z})} & z \in L \cap -\mathbb{H}. \end{cases}$$

The proof that Scharz reflection gives a holomorphic function also proves that this function is holomorphic as a function $\mathbb{C} \setminus \{z = 0\} \rightarrow \mathbb{C}$.

We can translate G into a holomorphic extension of f by conjugating $h^{-1} \circ G \circ h : \mathbb{C} \setminus \{p\} \rightarrow \mathbb{C}$ where p is the preimage of 1 under the continuous extension $\bar{f}$.

If f has more than one point on S^1 which is sent to 1 (i.e. more than one point on $\mathbb{R}$ is sent to ∞ under h) then we can repeat a similar procedure, except now we Schwarz reflect vertical strips bounded by the vertical lines through the points on the real line which are sent to ∞ by h .

Lastly, must there only be finitely many points?

■

3

■

4

■

4

■